# What is “Responsible Data Science”?

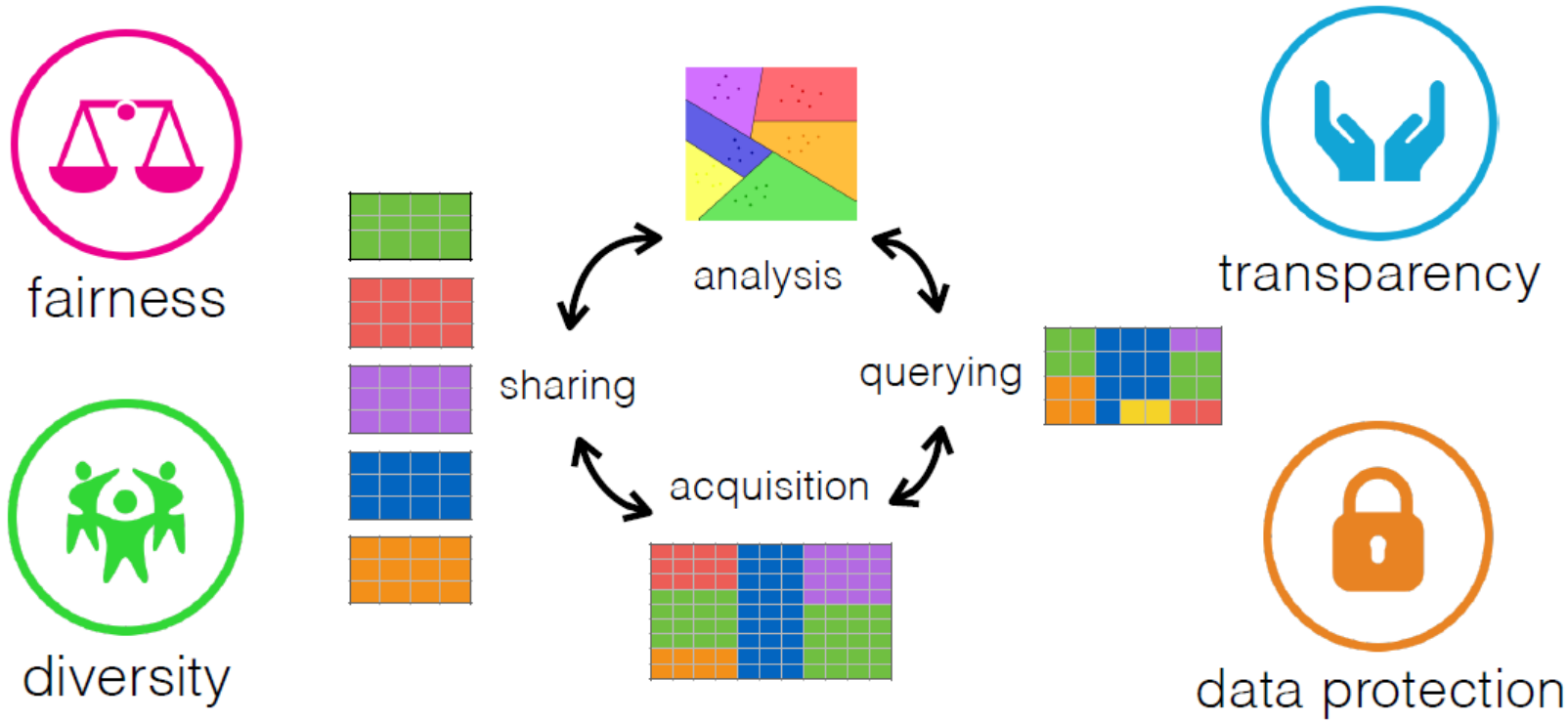
Completely based on the following NYU course:

DS-GA 3001.009: Special Topics in Data Science: Responsible Data Science

By Julia Stoyanovich, Associate Professor of Data Science

https://dataresponsibly.github.io/

# Responsible Data Science

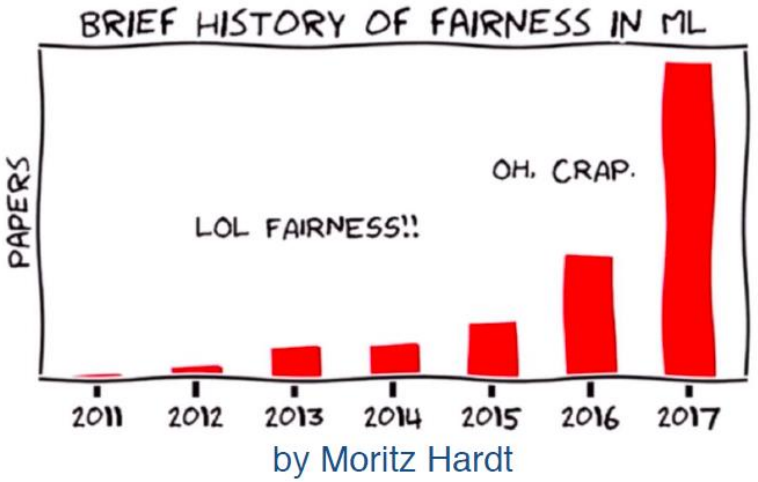



"Because of its power, data science should be used responsibly"

# Fairness

Works directly related to fairness in ML

- ICML 2020: 17 papers identify themselves under “Fairness, Equity, Justice, and Safety Subject
- NeurIPS 2019: 20 papers with the term “fair” in their titles
- AAAI 2020: 20 papers with the term “fair” in their titles
- …



Disclaimer: The following examples are mostly taken from news outlets and might not be the result of a rigorous scientific study

## Example: Online price discrimination

THE WALL STREET JOURNAL.

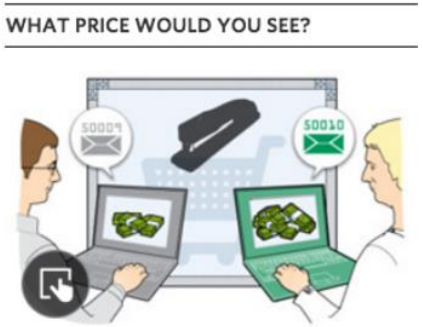
WHAT THEY KNOW

### Websites Vary Prices, Deals Based on Users' Information

By JENNIFER VALENTINO-DEVRIES, JEREMY SINGER-VINE and ASHKAN SOLTANI
December 24, 2012

It was the same Swingline stapler, on the same Staples.com website. But for Kim Wamble, the price was $15.79, while the price on Trude Frizzell's screen, just a few miles away, was $14.29.

A key difference: where Staples seemed to think they were located.




**lower prices offered to buyers who live in more affluent neighborhoods**

[https://www.wsj.com/articles/SB10001424127887323777204578189391813881534]
[https://lifehacker.com/how-web-sites-vary-prices-based-on-your-information-an-5973689]

## Example: Online job ads discrimination

"In particular, we found that males were shown ads encouraging the seeking of coaching services for high paying jobs more than females."

the fake users started with completely fresh profiles and behaved in the same way, with gender being the only factor that was different



theguardian

Samuel Gibbs

Wednesday 8 July 2015 11.29 BST

Women less likely to be shown ads for high-paid jobs on Google, study shows

Automated testing and analysis of company's advertising system reveals male job seekers are shown far more adverts for high-paying executive jobs

One experiment showed that Google displayed adverts for a career coaching service for executive jobs 1,852 times to the male group and only 318 times to the female group. Photograph: Alamy

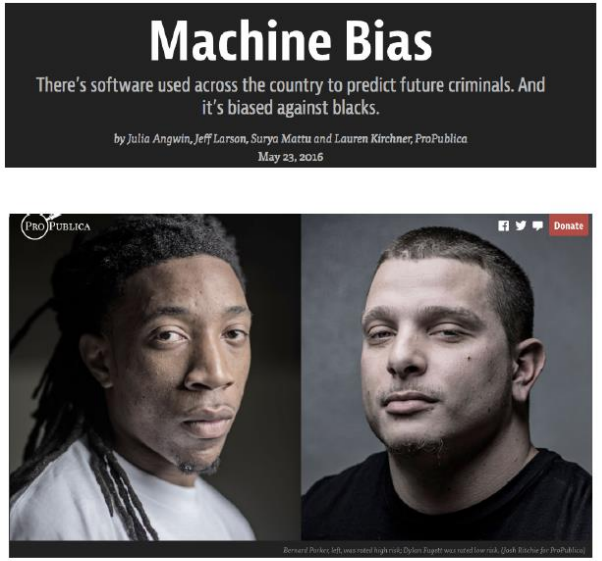
The AdFisher tool simulated job seekers that did not differ in browsing behavior, preferences or demographic characteristics, except in gender.

One experiment showed that Google displayed ads for a career coaching service for "$200k+" executive jobs **1,852 times to the male group and only 318 times to the female group**. Another experiment, in July 2014, showed a similar trend but was not statistically significant.

[https://www.theguardian.com/technology/2015/jul/08/women-less-likely-ads-high-paid-jobs-google-study]

## Example: Racial Discrimination

"Prater was the more seasoned criminal. He had already been convicted of armed robbery and attempted armed robbery, for which he served five years in prison, in addition to another armed robbery charge. Borden had a record, too, but it was for misdemeanors committed when she was a juvenile.
Yet something odd happened when Borden and Prater were booked into jail: A computer program spat out a score predicting the likelihood of each committing a future crime. Borden — who is black — was rated a high risk. Prater — who is white — was rated a low risk."




A commercial tool COMPAS automatically predicts some categories of future crime to assist in bail and sentencing decisions. It is used in courts in the US.

The tool correctly predicts recidivism **61% of the time.**

**Blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend.**

The tool makes **the opposite mistake among whites**: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes.

[https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing]

[Related paper: Corbett et al. "Algorithmic decision making and the cost of fairness"

# Example: Twitter cropping tool

Automatically Cropped Thumbnail




Original Image





[https://www.theguardian.com/technology/2020/sep/21/twitter-apologises-for-racist-image-cropping-algorithm]

# Biased ML

**“Definition”**

A dataset, model or algorithm that does not represent the world in a fair or correct manner

## Reflection/Discussion point

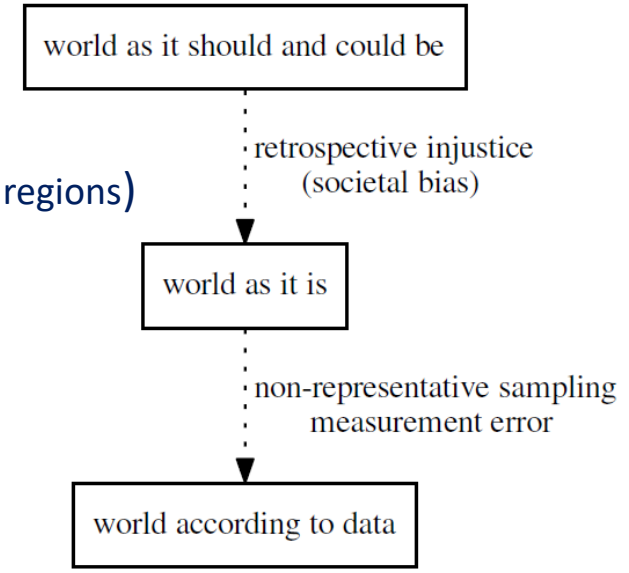
- What are the potential causes for unfair or incorrect model or dataset or algorithm?



3 minutes

# Why do/can we have biased ML?

- Biased Data
  - data is not representative of the source
    - bias in data collection (e.g., only from Europe)
    - bias in data cleaning (e.g., remove low-quality data; coming from poor regions)
    - measurement error (e.g., different collection mechanisms)
    - …
    - but it is not necessarily statistical bias!
  - source (world) is biased
    - *"the world as it is or as it should be?"*



## Why do/can we have biased ML?



- Biased Data
- Biased Model
  - e.g., ignores the long tail
- Biased Learning Process
  - e.g., the training procedure underweights the underrepresented classes
- Correlation for causality?
  - we are looking for causality, our models learns the correlations
- Social Biases
  - Reflected in semantics, problem formulations, tasks, etc.



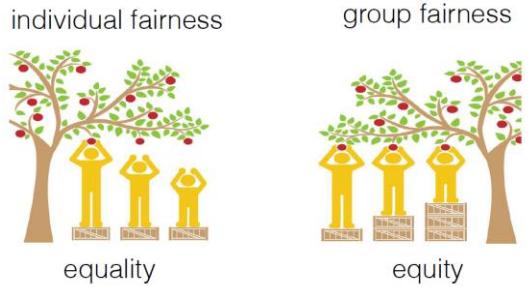

[Ferenc Huszár Twitter]

when data/model/algorithm involves people,

bias can lead to discrimination

# Biased ML

## How to avoid it? (open research question)




- Transparency
  - Model Interpretability
  - Decision Explainability
- Learn causality
- Human in the loop
- Systematically test for discrimination, add a discrimination loss!
- Removing (decorrelating) discrimination features (protected attributes)
  - However remember! Mere removal does not solve the problem! (other “safe” features might/most likely correlate e.g. race/zipcode)
- Education
- …

**Interested?**

Visit the following course page:

https://dataresponsibly.github.io/